gpeyt1-42 · Strip footing on clay (φ = 0)

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A 5 ft wide continuous wall footing bears 3.5 ft deep in a saturated, homogeneous stiff clay. Unconfined compression tests give an average qu = 2,400 psf; γ = 118 pcf; the water table is deep. Analyze the short-term (undrained) case with FS = 3.0 on net bearing pressure. The wall load is 9 kips/ft.

Most nearly, the allowable NET bearing pressure is:

A. 4,112 psf
B. 2,194 psf
C. 2,280 psf
D. 2,056 psf

Given equations / data

> φ = 0: Nc = 5.14, Nq = 1, Nγ = 0; su = qu/2; qnet,ult = qult − γDf = 5.14 su

Worked solution

1. su = 2,400/2 = 1,200 psf
2. qnet,ult = 5.14 × 1,200 = 6,168 psf
3. qnet,all = 6,168/3.0 = 2,056 psf
4. (check: wall pressure 9,000/5 = 1,800 psf gross)

Final answer

2,056 psf

Why the other values are wrong

- **4,112 psf** — used qu as su (forgot ÷2)
- **2,194 psf** — reported gross (added γDf after FS)
- **2,280 psf** — used Terzaghi Nc = 5.7

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.4.2 Bearing Capacity Equation (strip) and factor table (φ = 0: Nc = 5.14), pp. 90–91; §3.3.3 Undrained Shear Strength

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=l9jr9yfBzG0, https://www.youtube.com/watch?v=ZUGdBao3tkc (no creator values copied).

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